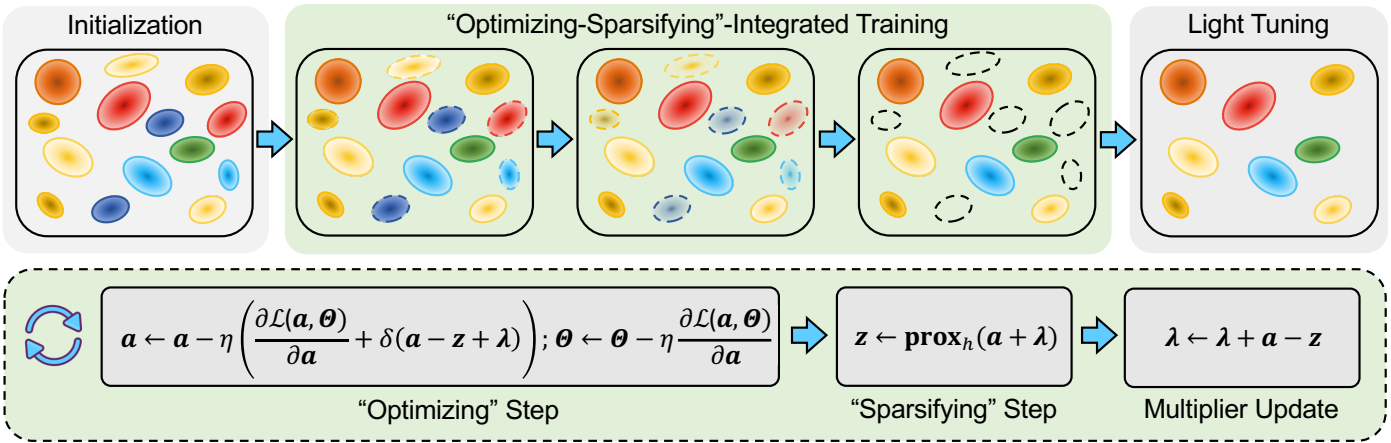
Initialization
"Optimizing-Sparsifying"-Integrated Training
Light Tuning
$\boldsymbol{a} \leftarrow \boldsymbol{a} - \eta \left(\frac{\partial \mathcal{L}(\boldsymbol{a}, \boldsymbol{\Theta})}{\partial \boldsymbol{a}} + \delta(\boldsymbol{a} - \boldsymbol{z} + \boldsymbol{\lambda}) \right); \boldsymbol{\Theta} \leftarrow \boldsymbol{\Theta} - \eta \frac{\partial \mathcal{L}(\boldsymbol{a}, \boldsymbol{\Theta})}{\partial \boldsymbol{a}}$
"Optimizing" Step
$\boldsymbol{z} \leftarrow \mathbf{prox}_h(\boldsymbol{a} + \boldsymbol{\lambda})$
"Sparsifying" Step
$\boldsymbol{\lambda} \leftarrow \boldsymbol{\lambda} + \boldsymbol{a} - \boldsymbol{z}$
Multiplier Update